

Math 2320 - FE 1.1 Review

1. $\int \frac{x}{x^2 - 1} dx$

2. $\int \frac{x^3}{x^2 - 1} dx$

3. $\int \frac{\ln(x)}{x} dx$

4. $\int x \ln(x) dx$

5. $\int x e^{x^2} dx$

6. $\int x^2 e^x dx$

7. $\int \arctan(x) dx$

8. What are the radii and intervals of convergence of the following power series.

(a) $\sum_{n=1}^{\infty} \frac{1}{n} x^n$

(b) $\sum_{n=1}^{\infty} \frac{1}{n^2} x^n$

(c) $\sum_{n=1}^{\infty} \frac{1}{n^2} (x-5)^n$

(d) $\sum_{n=0}^{\infty} \frac{1}{n!} x^n$

(e) $\sum_{n=0}^{\infty} x^n$

(f) $\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}$

9. Find the Taylor series from the definition

(a) $f(x) = \sin(2x)$ at $x = 0$

(b) $f(x) = e^x$ at $x = 0$

(c) $f(x) = \frac{1}{1-x}$ at $x = 0$

(d) $f(x) = \ln(x+1)$ at $x = 0$